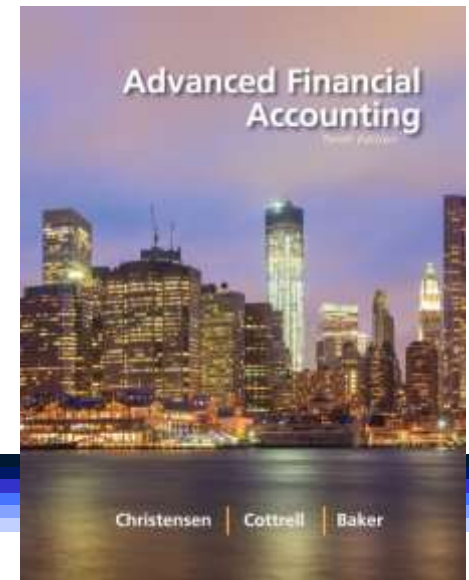


Chapter 8



Intercompany Indebtedness



Learning Objective 8-1

Understand and explain concepts associated with intercompany debt transfers.



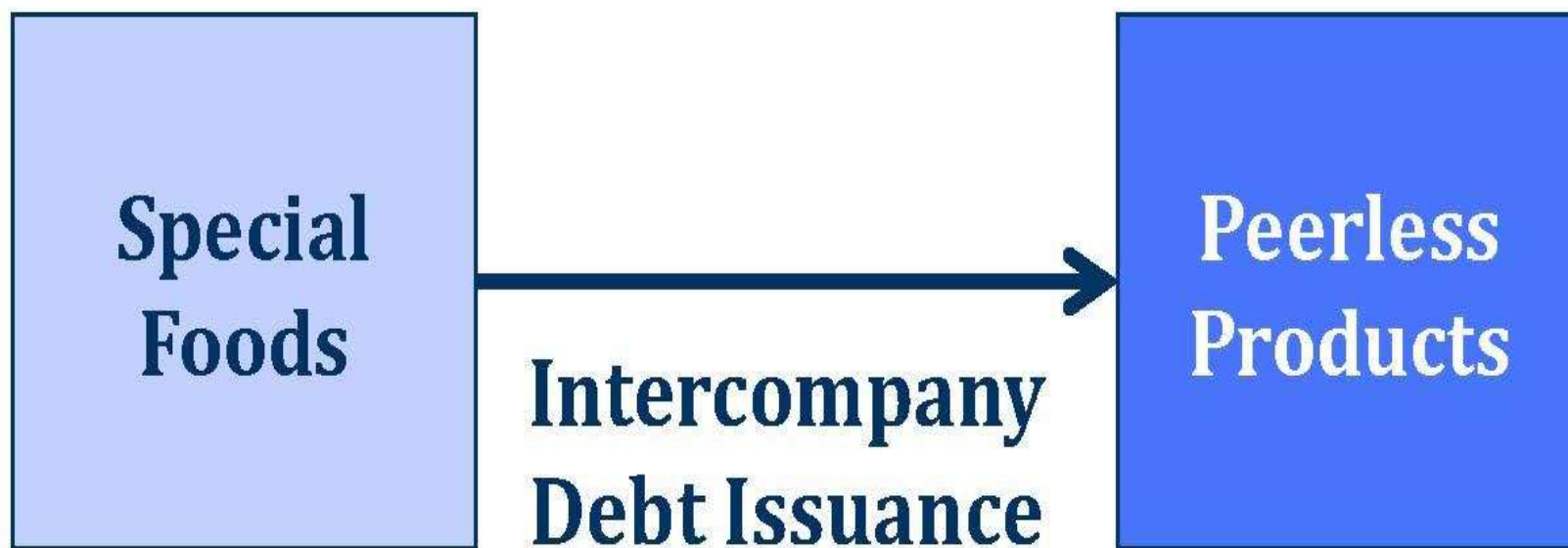
Consolidation Overview

- ◆ **A direct intercompany debt transfer involves a loan from one affiliate to another without the participation of an unrelated party.**
- ◆ **An indirect intercompany debt transfer involves the issuance of debt to an unrelated party and the subsequent purchase of the debt instrument by an affiliate of the issuer.**



Direct Intercompany Debt Transfer

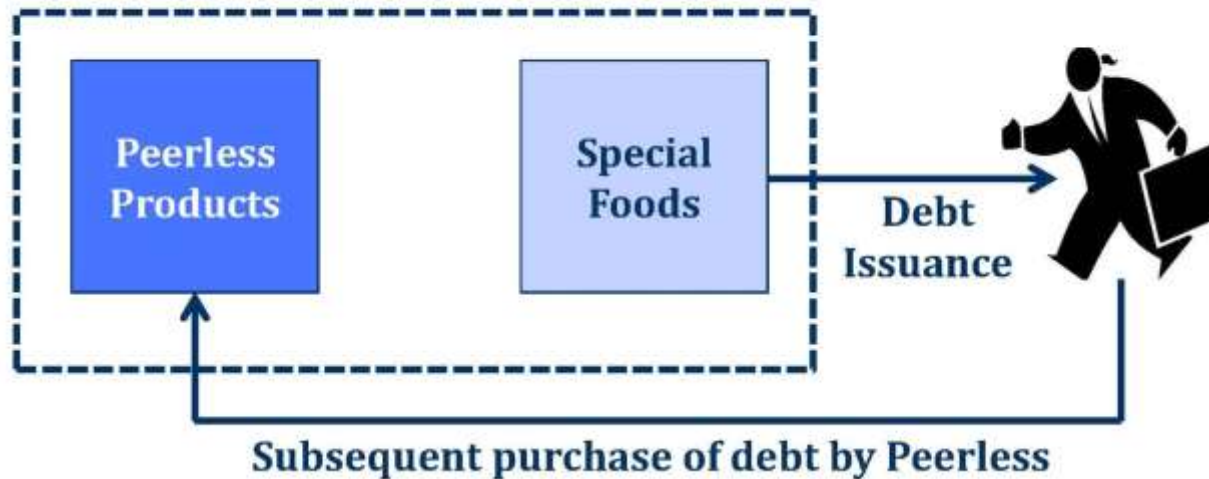
(a) Direct Intercompany Debt Transfer





Indirect Intercompany Debt Transfer

(b) Indirect Intercompany Debt Transfer





Practice Quiz Question #1

Which of the following statements is true?

- a. A direct intercompany debt transfer always involves an unrelated party.
- b. An indirect intercompany debt transfer always involves a transfer directly to a related party.
- c. A direct intercompany debt transfer never involves an unrelated party.
- d. An indirect intercompany debt transfer is first transferred to an affiliated company with a subsequent transfer to an unrelated



Practice Quiz Question #1 *Solution*

Which of the following statements is true?

- a. A direct intercompany debt transfer always involves an unrelated party.
- b. An indirect intercompany debt transfer always involves a transfer directly to a related party.
- c. A direct intercompany debt transfer never involves an unrelated party.
- d. An indirect intercompany debt transfer is first transferred to an affiliated company with a subsequent transfer to an unrelated



Learning Objective 8-2

Prepare journal entries and elimination entries related to direct intercompany debt transfers.



Bond Sale Directly to an Affiliate

- ◆ **When one company sells bonds directly to an affiliate, all effects of the intercompany indebtedness must be eliminated in preparing consolidated financial statements.**
- ◆ **Transfer at par value**
(example continues on next slide).



Bond Sale Directly to an Affiliate

Assume that on January 1, 20X1, Special Foods borrows \$100,000 from Peerless Products by issuing \$100,000 par value, 12 percent, 10-year bonds. During 20X1, Special Foods records interest expense on the bonds of \$12,000 ($\$100,000 \times .12$), and Peerless records an equal amount of interest income.

In the preparation of consolidated financial statements for 20X1, two elimination entries are needed in the consolidation worksheet to remove the effects of the intercompany indebtedness:



Bond Sale Directly to an Affiliate

Assume that on January 1, 20X1, Special Foods borrows \$100,000 from Peerless Products by issuing \$100,000 par value, 12 percent, 10-year bonds. During 20X1, Special Foods records interest expense on the bonds of \$12,000 ($\$100,000 \times .12$), and Peerless records an equal amount of interest income.

In the preparation of consolidated financial statements for 20X1, two elimination entries are needed in the consolidation worksheet to remove the effects of the intercompany indebtedness:

Eliminate intercorporate bond holding:

Bonds Payable	100,000	
Investment in Special Foods Bonds		100,000

Eliminate intercompany interest:

Interest Income	12,000	
Interest Expense		12,000



Bond Sale Directly to an Affiliate

◆ Transfer at a discount or premium

- Bond interest income or expense recorded do not equal cash interest payments.
- Interest income/expense amounts are adjusted for the amortization of the discount or premium.



Bond Sale Directly to an Affiliate

On January 1, 20X1, Peerless Products purchases \$100,000 par value, 12 percent, 10-year bonds from Special Foods when the market interest rate is 13 percent. In order to yield a 13 percent return, Special Foods issues the bonds at a discount for \$94,490.75. Interest on the bonds is payable on January 1 and July 1.

date	interest payment	interest expense	amort.	discount	carrying value



Bond Sale Directly to an Affiliate

On January 1, 20X1, Peerless Products purchases \$100,000 par value, 12 percent, 10-year bonds from Special Foods when the market interest rate is 13 percent. In order to yield a 13 percent return, Special Foods issues the bonds at a discount for \$94,490.75. Interest on the bonds is payable on January 1 and July 1.

date	interest payment	interest expense	amort.	discount	face value
Jan 1, 20X1				5,509.25	100,000
Jul 1, 20X1					



Bond Sale Directly to an Affiliate

On January 1, 20X1, Peerless Products purchases \$100,000 par value, 12 percent, 10-year bonds from Special Foods when the market interest rate is 13 percent. In order to yield a 13 percent return, Special Foods issues the bonds at a discount for \$94,490.75. Interest on the bonds is payable on January 1 and July 1.

date	interest payment	interest expense	amort.	discount	carrying value
Jan 1, 20X1				5,509.25	94,490.75



Bond Sale Directly to an Affiliate

On January 1, 20X1, Peerless Products purchases \$100,000 par value, 12 percent, 10-year bonds from Special Foods when the market interest rate is 13 percent. In order to yield a 13 percent return, Special Foods issues the bonds at a discount for \$94,490.75. Interest on the bonds is payable on January 1 and July 1.

date	interest payment	interest expense	amort.	discount	carrying value
Jan 1, 20X1				5,509.25	94,490.75
Jul 1, 20X1	6000	6,141.90	141.90	5,367.35	94,632.65



Bond Sale Directly to an Affiliate

On January 1, 20X1, Peerless Products purchases \$100,000 par value, 12 percent, 10-year bonds from Special Foods when the market interest rate is 13 percent. In order to yield a 13 percent return, Special Foods issues the bonds at a discount for \$94,490.75. Interest on the bonds is payable on January 1 and July 1.

date	interest payment	interest expense	amort.	discount	carrying value
Jan 1, 20X1				5,509.25	94,490.75
Jul 1, 20X1	6000	6,141.90	141.90	5,367.35	94,632.65
jan 1, 20X2	6000	6,151.12	151.12	5,216.23	94,783.77



Bond Sale Directly to an Affiliate

On January 1, 20X1, Peerless Products purchases \$100,000 par value, 12 percent, 10-year bonds from Special Foods when the market interest rate is 13 percent. In order to yield a 13 percent return, Special Foods issues the bonds at a discount for \$94,490.75. Interest on the bonds is payable on January 1 and July 1.

date	interest payment	interest expense	amort.	discount	carrying value
Jan 1, 20X1				5,509.25	94,490.75
Jul 1, 20X1	6000	6,141.90	141.90	5,367.35	94,632.65
jan 1, 20X2	6000	6,151.12	151.12	5,216.23	94,783.77
Jul 1, 20X2	6000	6,160.95	160.95	5,055.28	94,944.72



Bond Sale Directly to an Affiliate

On January 1, 20X1, Peerless Products purchases \$100,000 par value, 12 percent, 10-year bonds from Special Foods when the market interest rate is 13 percent. In order to yield a 13 percent return, Special Foods issues the bonds at a discount for \$94,490.75. Interest on the bonds is payable on January 1 and July 1. The interest expense recognized by Special Foods and the interest income recognized by Peerless each period based on effective interest amortization of the discount over the life of the bonds can be summarized as follows:

Bond Discount Amortization Table

Payment Number	Period End	Interest Payment (Face x 0.12 x 6/12)	Interest Expense (Carrying Value x 0.13 x 6/12)	Amortization of Discount	Discount	Bonds Payable Face Value	Carrying Value of Bonds
					(5,509.25)	100,000	94,490.75
1	7/1/20X1	6,000	6,141.90	141.90	(5,367.36)	100,000	94,632.64
2	1/1/20X2	6,000	6,151.12	151.12	(5,216.23)	100,000	94,783.77
3	7/1/20X2	6,000	6,160.94	160.94	(5,055.29)	100,000	94,944.71
4	1/1/20X3	6,000	6,171.41	171.41	(4,883.88)	100,000	95,116.12
5	7/1/20X3	6,000	6,182.55	182.55	(4,701.33)	100,000	95,298.67
6	1/1/20X4	6,000	6,194.41	194.41	(4,506.92)	100,000	95,493.08
7	7/1/20X4	6,000	6,207.05	207.05	(4,299.87)	100,000	95,700.13



Bond Sale Directly to an Affiliate

◆ Entries by the debtor

January 1, 20X1

Cash	94,491	
	Bonds Payable	94,491

Issue bonds to Peerless Products.

July 1, 20X1

Interest Expense	6,142	
	Bonds Payable	142
	Cash	6,000

Semiannual payment of interest.

December 31, 20X1

Interest Expense	6,151	
	Bonds Payable	151
	Interest Payable	6,000

Accrue interest expense at year-end.



Bond Sale Directly to an Affiliate

◆ Entries by the bond investor

January 1, 20X1

Investment in Special Foods Bonds	94,491
Cash	94,491

Purchase of bonds from Special Foods.

July 1, 20X1

Cash	6,000	
Investment in Special Foods Bonds	142	
Interest Income		6,142

Receive interest on bond investment

December 31, 20X1

Interest Receivable	6,000	
Investment in Special Foods Bonds	151	
Interest Income		6,151

Accrue interest income at year-end.



Bond Sale Directly to an Affiliate

Elimination Entries at Year-End—20X1:

Eliminates the bonds payable and associated discount against the investment in bonds:

Bonds Payable	94,784	
Investment in Special Foods Bonds		94,784

Eliminates the bond interest income recognized by Peerless during 20X1 against the bond interest expense recognized by Special Foods:

Interest Income	12,293	
Interest Expense		12,293

Eliminates the interest receivable against the interest payable.

Interest Payable	6,000	
Interest Receivable		6,000



Bond Sale Directly to an Affiliate

Elimination Entries at Year-End—20X2:

Eliminates intercorporate bond holding.

Bonds Payable	95,116	
Investment in Special Foods Bonds		95,116

Eliminates intercompany interest:

Interest Income	12,332	
Interest Expense		12,332

Eliminates intercompany interest receivable/payable.

Interest Payable	6,000	
Interest Receivable		6,000

Consolidation at the end of 20X2 requires elimination entries similar to those at the end of 20X1. By the end of the second year (i.e., the fourth interest payment), the carrying value of the bond investment on Peerless' books increases to \$95,116 (\$94,491 issue price + discount amortization of \$142 + \$151 + \$161 + 171). Similarly, the bond discount on Special Foods' books decreases to \$4,884, resulting in an effective bond liability of \$95,116.



Bonds of Affiliate Purchased from a Nonaffiliate

- ◆ ***Scenario:* Bonds that were issued to an unrelated party are acquired later by an affiliate of the issuer.**
 - From the viewpoint of the consolidated entity, an acquisition of an affiliate's bonds retires the bonds at the time they are purchased.
- ◆ **Acquisition of the bonds of an affiliate by another company within the consolidated entity is referred to as constructive retirement.**
 - Although the bonds actually are not retired, they are treated as if they were retired in preparing consolidated financial statements.



Bonds of Affiliate Purchased from a Nonaffiliate

- ◆ **When a constructive retirement occurs:**
 - The consolidated income statement for the period reports a gain or loss on debt retirement based on the difference between the carrying value of the bonds on the books of the debtor and the purchase price paid by the affiliate.
 - Neither the bonds payable nor the purchaser's investment in the bonds is reported in the consolidated balance sheet because the bonds are no longer considered outstanding.



Practice Quiz Question #2

A constructive debt retirement is

- a. an acquisition of the bonds of an affiliate by another company within the consolidated entity.
- b. an acquisition of the bonds of an non-affiliate by a company within a consolidated entity.
- c. a sale of the bonds of a non-affiliate by a company within a consolidated entity.
- d. a sale of the bonds of an affiliate to a company within a consolidated entity.



Practice Quiz Question #2 *Solution*

A constructive debt retirement is

- a. an acquisition of the bonds of an affiliate by another company within the consolidated entity.
- b. an acquisition of the bonds of an non-affiliate by a company within a consolidated entity.
- c. a sale of the bonds of a non-affiliate by a company within a consolidated entity.
- d. a sale of the bonds of an affiliate to a company within a consolidated entity.



Learning Objective 8-3

Prepare journal entries and elimination entries related to debt purchased from a nonaffiliate to an amount less than book value.



Bonds of Affiliate Purchased from a Nonaffiliate

- ◆ **Purchase at an amount less than book value:**
 - When the price paid to acquire the bonds of an affiliate differs from the liability reported by the debtor, a gain or loss is reported in the consolidated income statement in the period of constructive retirement.
 - The bond interest income and interest expense reported by the two affiliates subsequent to the purchase must be eliminated in preparing consolidated statements.
 - Interest income reported by the investing affiliate and interest expense reported by the debtor are not equal in this case because of the different bond carrying amounts on the books of the two companies.



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration

Peerless Products Corporation acquires 80 percent of the common stock of Special Foods Inc. on December 31, 20X0, for its underlying book value of \$240,000. At that date, the fair value of the noncontrolling interest is equal to its book value of \$60,000. Additionally:

1. On January 1, 20X1, Special Foods issues 10-year, 12 percent bonds payable with a par value of \$100,000; the bonds are issued at a premium, \$105,975.19, to yield the current market interest rate of 11%. Nonaffiliated Corporation purchases the bonds from Special Foods.
2. The bonds pay interest on June 30 and December 31.
3. On December 31, 20X1, Peerless purchases the bonds from Nonaffiliated for \$94,823.04 when the bonds' carrying value on Special's books is \$105,623.04, resulting in a gain of \$10,800 on the constructive retirement of the bonds. Note that Peerless' purchase price reflects the current market interest rate of 12.992186% when the bonds have 18 payments left to maturity.
4. Special Foods reports net income of \$50,152 for 20X1 and \$75,192 for 20X2 and declares dividends of \$30,000 in 20X1 and \$40,000 in 20X2.
5. Peerless earns \$140,000 in 20X1 and \$160,000 in 20X2 from its own separate operations. Peerless declares dividends of \$60,000 in both 20X1 and 20X2.



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration

Peerless Products Corporation acquires 80 percent of the common stock of Special Foods Inc. on December 31, 20X0, for its underlying book value of \$240,000. At that date, the fair value of the noncontrolling interest is equal to its book value of \$60,000. Additionally:

1. On January 1, 20X1, Special Foods issues 10-year, 12 percent bonds payable with a par value of \$100,000; the bonds are issued at a premium, \$105,975.19, to yield the current market interest rate of 11%. Nonaffiliated Corporation purchases the bonds from Special Foods.
2. The bonds pay interest on June 30 and December 31.

date	interest payment	interest expense	amort.	Premium	carrying value
Jan 1, 20X1				5,975.19	105,975.19
Jun 30, 20X1					



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration

1. On January 1, 20X1, Special Foods issues 10-year, 12 percent bonds payable with a par value of \$100,000; the bonds are issued at a premium, \$105,975.19, to yield the current market interest rate of 11%. Nonaffiliated Corporation purchases the bonds from Special Foods.
2. The bonds pay interest on June 30 and December 31.

date	interest payment	interest expense	amort.	Premium	carrying value
Jan 1, 20X1				5,975.19	105,975.19
Jun 30, 20X1	6000	5,828.64	171.36	5,803.83	105,803.83



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration

1. On January 1, 20X1, Special Foods issues 10-year, 12 percent bonds payable with a par value of \$100,000; the bonds are issued at a premium, \$105,975.19, to yield the current market interest rate of 11%. Nonaffiliated Corporation purchases the bonds from Special Foods.
2. The bonds pay interest on June 30 and December 31.

date	interest payment	interest expense	amort.	Premium	carrying value
Jan 1, 20X1				5,975.19	105,975.19
Jun 30, 20X1	6000	5,828.64	171.36	5,803.83	105,803.83
Dec 31, 20X1	6000	5,819.21	180.79	5,623.04	105,623.04

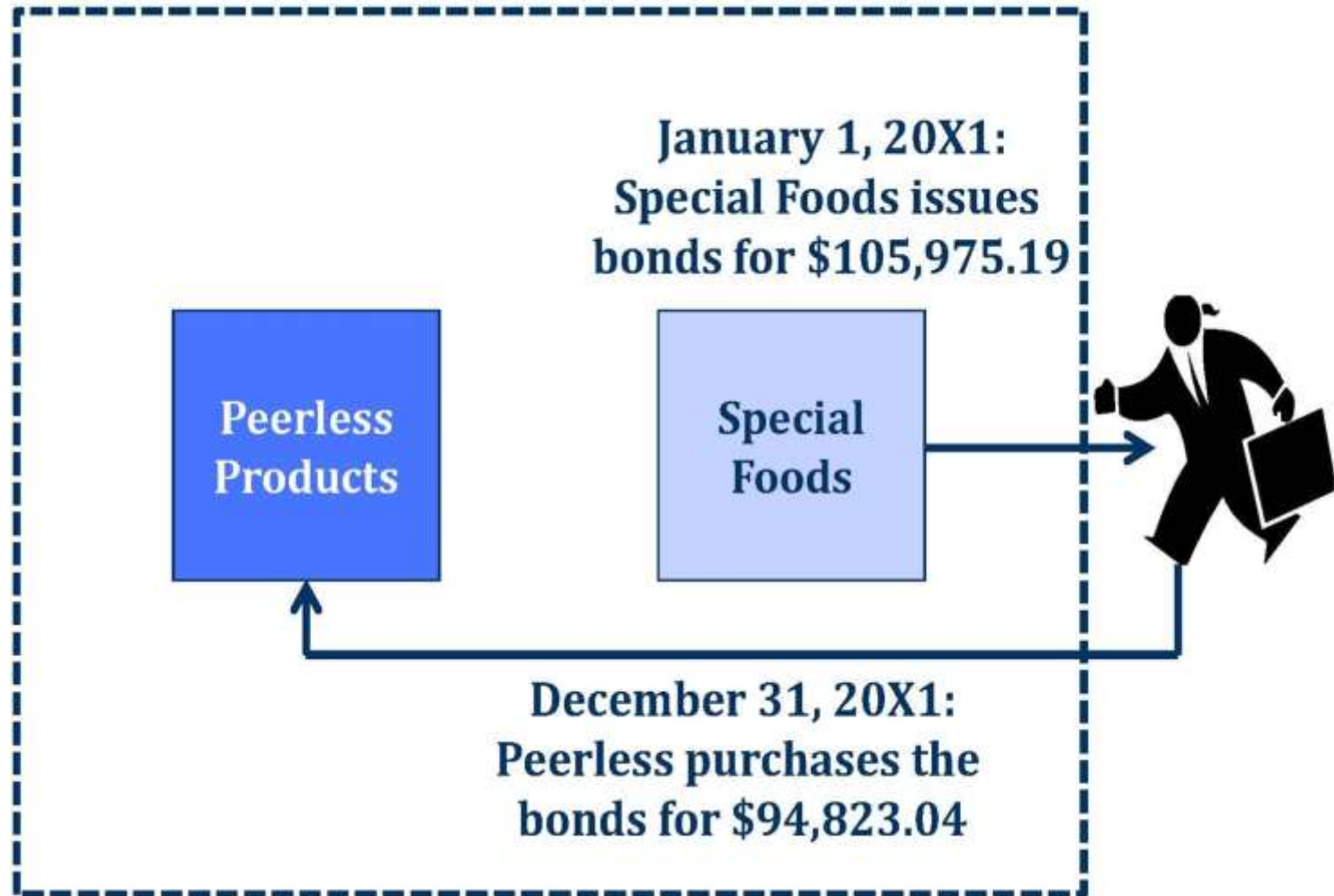


Bonds of Affiliate Purchased from a Nonaffiliate: Illustration

3. On December 31, 20X1, Peerless purchases the bonds from Nonaffiliated for \$94,823.04 when the bonds' carrying value on Special's books is \$105,623.04, resulting in a gain of \$10,800 on the constructive retirement of the bonds. Note that Peerless' purchase price reflects the current market interest rate of 12.992186% when the bonds have 18 payments left to maturity.



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration





Bond Sale Directly to an Affiliate

In addition, the interest expense recognized by Special Foods each period based on the effective interest amortization of the premium over the life of the bonds can be summarized as follows:

Bond Premium Amortization Table							
Payment Number	Period End	Interest Payment (Face x 0.12 x 6/12)	Interest Expense (Carrying Value x 0.11 x 6/12)	Amortization of Premium	Premium	Bonds Payable Face Value	Carrying Value of Bonds
	1/1/20X1						105,975.19
1	6/30/20X1	6,000.00	5,828.64	(171.36)	5,803.83	100,000.00	105,803.83
2	12/31/20X1	6,000.00	5,819.21	(180.79)	5,623.04	100,000.00	105,623.04
3	6/30/20X2	6,000.00	5,809.27	(190.73)	5,432.30	100,000.00	105,432.30
4	12/31/20X2	6,000.00	5,798.78	(201.22)	5,231.08	100,000.00	105,231.08
5	6/30/20X3	6,000.00	5,787.71	(212.29)	5,018.79	100,000.00	105,018.79
6	12/31/20X3	6,000.00	5,776.03	(223.97)	4,794.82	100,000.00	104,794.82
7	6/30/20X4	6,000.00	5,763.72	(236.28)	4,558.54	100,000.00	104,558.54
8	12/31/20X4	6,000.00	5,750.72	(249.28)	4,309.26	100,000.00	104,309.26
9	6/30/20X5	6,000.00	5,737.01	(262.99)	4,046.27	100,000.00	104,046.27
10	12/31/20X5	6,000.00	5,722.54	(277.46)	3,768.81	100,000.00	103,768.81
11	6/30/20X6	6,000.00	5,707.28	(292.72)	3,476.10	100,000.00	103,476.10
12	12/31/20X6	6,000.00	5,691.19	(308.81)	3,167.28	100,000.00	103,167.28
13	6/30/20X7	6,000.00	5,674.20	(325.80)	2,841.48	100,000.00	102,841.48



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration—YEAR 1

Bond Liability Entries—20X1 (Special Foods)

January 1, 20X1

Cash	105,975	
Bonds Payable		105,975

Sale of bonds to Nonaffiliated.

June 30, 20X1

Interest Expense	5,829	
Bonds Payable	171	
	Cash	6,000

Semiannual payment of interest.

December 31, 20X1

Interest Expense	5,819	
Bonds Payable	181	
	Cash	6,500

Semiannual payment of interest.



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration—YEAR 1

Total interest expense for 20X1 is \$11,648 (\$5,829 + \$5,819), and the book value of the bonds on December 31, 20X1, is \$105,623 as shown in the amortization table.

Bond Premium Amortization Table							
Payment Number	Period End	Interest Payment (Face x 0.12 x 6/12)	Interest Expense (Carrying Value x 0.11 x 6/12)	Amortization of Premium	Premium	Bonds Payable Face Value	Carrying Value of Bonds
	1/1/20X1						105,975.19
1	6/30/20X1	6,000.00	5,828.64	(171.36)	5,803.83	100,000.00	105,803.83
2	12/31/20X1	6,000.00	5,819.21	(180.79)	5,623.04	100,000.00	105,623.04

Bond Investment Entry—20X1 (Peerless)

December 13, 20X1

Investment in Special Foods Bonds	94,823
Cash	94,823

Purchase of Special Foods bonds from Nonaffiliated Corporation.

Computation of Gain on Constructive Retirement of Bonds

Book value of Special Foods' bonds, December 31, 20X1	\$105,623
Price paid by Peerless to purchase bonds (94,823)	
Gain on constructive retirement of bonds	\$10,800

This gain is included in the consolidated income statement as a gain on the retirement of bonds.



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration

◆ Assignment of gain: constructive retirement

■ Four approaches have been used:

- 1.** The affiliate issuing the bonds
- 2.** The affiliate purchasing the bonds
- 3.** The parent company
- 4.** The issuing and purchasing companies, based on the difference between the carrying amounts of the bonds on their books at the date of purchase and the par value of the bonds



Assignment of gain/loss from constructive retirement

- ◆ When the subsidiary is the issuing affiliate, the gain/loss on constructive retirement is viewed as accruing to the subsidiary's shareholders. Thus, the gain or loss is apportioned between the controlling and non-controlling interest based on relative ownership
- ◆ If the parent is the issuing affiliate, the gain/loss accrues to the controlling interest and none is apportioned to the non-controlling interest.



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration—YEAR 1

Fully Adjusted Equity-Method Entries—20X1: In addition to recording the bond investment, Peerless records the following equity-method entries during 20X1 to account for its investment in Special Foods stock:

Investment in Special Foods Stock	40,122	
Income from Special Foods		40,122

Record equity-method income: \$50,152 x 0.80.

Cash	24,000	
Investment in Special Foods Stock		24,000

Record dividends from Special Foods: \$30,000 x 0.80.

Investment in Special Foods Stock	8,640	
Income from Special Foods		8,640

Record Peerless' 80% share of the gain on the constructive retirement of Special Foods' bonds.



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration—YEAR 1

These entries result in a \$264,640 balance in the investment account at the end of 20X1 as shown:

Investment in Special Foods		Income from Special Foods	
Acq.	240,000		
80% NI	40,122		40,122 80% NI
	8,640	24,000 80% Dividends 80% of Bond Retirement Gain	8,640
EB	264,762		48,762 EB



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration—YEAR 1

$$48,762 = 40,122 + (80\% \times \text{constructive gain})$$

Book Value Calculations:

NCI 20%	Peerless 80%	Common Stock	Retained Earnings	=	+
Original Book Value	60,000	240,000	200,000	)	
+ Net Income	)	40,122	152		
- Dividends	(6,000)	(24,000)			
Ending Book Value	)	256,122	0,000	),152)	

Basic Investment Account Elimination Entry:

Common Stock	200,000	← Common Stock Balance
Retained Earnings	100,000	← Beg. RE from trial balance
Income from Special Foods	48,762	← Peerless' % of NI + 80% Ret. Gain
NCI in NI of Special Foods	12,190	← NCI share of NI + 20% Ret. Gain
Dividends Declared	30,000	← 100% of Sub's dividends
Investment in Special Foods Stock		← Net BV + 80% Ret. Gain
264,762		← NCI share of BV + 20% Ret. Gain



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration—YEAR 1

$$12,190 = 10,030 + (20\% \times \text{constructive gain})$$

Book Value Calculations:

NCI 20%	Peerless 80%	Common Stock	Retained Earnings	=	+
Original Book Value	60,000	240,000	200,000	)	
+ Net Income	)	40,122	152		
- Dividends	(6,000)	(24,000)			
Ending Book Value	)	256,122	0,000	(,152)	

Basic Investment Account Elimination Entry:

Common Stock	200,000	← Common Stock Balance
Retained Earnings	100,000	← Beg. RE from trial balance
Income from Special Foods	48,762	← Peerless' % of NI + 80% Ret. Gain
NCI in NI of Special Foods	12,190	← NCI share of NI + 20% Ret. Gain
Dividends Declared	30,000	← 100% of Sub's dividends
Investment in Special Foods Stock		← Net BV + 80% Ret. Gain
264,762		← NCI share of BV + 20% Ret. Gain



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration—YEAR 1

$$264,762 = 256,122 + (80\% \times \text{constructive gain})$$

Book Value Calculations:

	NCI 20%	Peerless 80%	Common Stock	Retained Earnings	=	+
Original Book Value		60,000	240,000	200,000	)	
+ Net Income		)	40,122	152		
- Dividends		(6,000)	(24,000)			
Ending Book Value	)		256,122	0,000	,152)	

Basic Investment Account Elimination Entry:

Common Stock	200,000	←	Common Stock Balance
Retained Earnings	100,000	←	Beg. RE from trial balance
Income from Special Foods	48,762	←	Peerless' % of NI + 80% Ret. Gain
NCI in NI of Special Foods	12,190	←	NCI share of NI + 20% Ret. Gain
Dividends Declared	30,000	←	100% of Sub's dividends
Investment in Special Foods Stock		←	Net BV + 80% Ret. Gain
264,762		←	NCI share of BV + 20% Ret. Gain



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration—YEAR 1

$$66,190 = 64,030 + (20\% \times \text{constructive gain})$$

Book Value Calculations:

	NCI 20%	Peerless 80%	Common Stock	+ Retained Earnings	=	+
Original Book Value		60,000	240,000	200,000	)	
+ Net Income		)	40,122		152	
- Dividends		(6,000)	(24,000)			
Ending Book Value	)		256,122	0,000	(,152)	

Basic Investment Account Elimination Entry:

Common Stock	200,000	←	Common Stock Balance
Retained Earnings	100,000	←	Beg. RE from trial balance
Income from Special Foods	48,762	←	Peerless' % of NI + 80% Ret. Gain
NCI in NI of Special Foods	12,190	←	NCI share of NI + 20% Ret. Gain
Dividends Declared	30,000	←	100% of Sub's dividends
Investment in Special Foods Stock		←	Net BV + 80% Ret. Gain
264,762		←	NCI share of BV + 20% Ret. Gain



Elimination entry—YEAR 1

(2) Worksheet entry to eliminate intercompany bond holdings:

Bonds Payable	105,623		
Investment in Special Foods Bonds		94,823	
Gain on Bond Retirement		10,800	

NCI 20%	Peerless 80%	Common Stock	+ Retained Earnings	=	+
Original Book Value	60,000	240,000	200,000	)	
+ Net Income	)	40,122		152	
- Dividends	(6,000)	(24,000)			
Ending Book Value	)	256,122	0,000	,152)	

(1) Basic Investment Account Elimination

Common Stock	200,000	← Common Stock Balance
Retained Earnings	100,000	← Beg. RE from trial balance
Income from Special Foods	48,762	← Peerless' % of NI + 80% Ret. Gain
NCI in NI of Special Foods	12,190	← NCI share of NI + 20% Ret. Gain
Dividends Declared	30,000	← 100% of Sub's dividends
Investment in Special Foods Stock		← Net BV + 80% Ret. Gain
264,762		← NCI share of BV + 20% Ret. Gain



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration—YEAR 1

The amounts related to the bonds from the books of Peerless and Special Foods and the appropriate consolidated amounts are shown below along with entry to eliminate intercompany bond holdings:

Item	Peerless Products	Special Foods	Unadjusted Totals	Consolidated Amounts
Bonds Payable	0	\$(105,623)	\$(105,623)	0
Investment in Bonds	\$94,823	0	94,823	0
Interest Expense	0	\$11,648	0	\$11,648
Interest Income	0	0	0	0
Gain on Bond Retirement	0	0	(10,800)	(10,800)

Worksheet entry to eliminate intercompany bond holdings:

Bonds Payable 105,623

Investment in Special Foods Bonds 94,823

Gain on Bond Retirement 10,800



Consolidation Worksheet at December 31, 20X2

	Peerless Products	Special Foods	Elimination Entries		Consolidated
			DR	CR	
Income Statement					
Sales	400,000	200,000			600,000
Less: COGS	(170,000)	(115,000)			(285,000)
Less: Depreciation Expense	(50,000)	(20,000)			(70,000)
Less: Other Expenses	(20,000)	(3,200)			(23,200)
Less: Interest Expense	(20,000)	(11,648)			(31,648)
Income from Special Foods	48,762		48,762		0
Gain from Bond Retirement				10,800	10,800
Consolidated Net Income	188,762	50,152	48,762	10,800	200,952
NCI in Net Income			12,190		(12,190)
Controlling Interest Net Income	188,762	50,152	60,952	10,800	188,762
Statement of Retained Earnings					
Beginning Balance	300,000	100,000	100,000		300,000
Net Income	188,762	50,152	60,952	10,800	188,762
Less: Dividends Declared	(60,000)	(30,000)		30,000	(60,000)
Ending Balance	428,762	120,152	160,952	40,800	428,762
Balance Sheet					
Cash	169,177	80,775			249,952
Accounts Receivable	75,000	50,000			125,000
Inventory	100,000	75,000			175,000
Investment in Special Foods Stock	264,762			264,762	0
Investment in Special Foods Bonds	94,823			94,823	0
Land	175,000	40,000			215,000
Building and Equipment	800,000	600,000		300,000	1,100,000
Less: Accumulated Depreciation	(450,000)	(320,000)	300,000		(470,000)
Total Assets	1,228,762	525,775	300,000	659,585	1,394,952
Accounts Payable	100,000	100,000			200,000
Bonds Payable	200,000	100,000	100,000		200,000
Premium on Bonds Payable		5,623	5,623		0
Common Stock	500,000	200,000	200,000		500,000
Retained Earnings	428,762	120,152	160,952	40,800	428,762
NCI in NA of Special Foods				66,190	66,190
Total Liabilities & Equity	1,228,762	525,775	466,775	106,990	1,394,952



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration—YEAR 1

◆ Consolidated Net Income—20X1

Peerless' separate income		140,000
Special Foods' net income	50,152	
Gain on constructive retirement of bonds	<u>10,800</u>	
Special Foods' realized net income		<u>60,952</u>
Consolidated net income, 20X1		200,952
Income to NCI= ($\$60,952 \times 0.2$)		<u>12,190</u>
Income to Controlling interest		<u>188,762</u>



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration—YEAR 1

◆ Noncontrolling Interest—December 31, 20X1

Book value of Special Foods' net assets, December 31, 20X1:

Common stock	\$200,000
Retained earnings	<u>120,152</u>
Total reported book value	\$320,152
Gain on constructive retirement of bonds	<u>10,800</u>
Realized book value of Special Foods' net assets	\$330,952
Noncontrolling stockholders' share	<u>× 0.20</u>
Noncontrolling interest in net assets, December 31, 20X1	<u>\$ 66,190</u>



Year 2

Special Foods amortization table

date	interest payment	interest expense	amort.	Premium	carrying value
Jan 1, 20X1				5,975	105,975
Jun 30, 20X1	6000	5,829	171	5,804	105,804
Dec 31, 20X1	6000	5,819	181	5,623	105,623
Jun 30, 20X2	6000	5,809	191	5,432	105,432
Dec 31, 20X2	6000	5,799	201	5,231	105,231



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration

3. On December 31, 20X1, Peerless purchases the bonds from Nonaffiliated for \$94,823.04 when the bonds' carrying value on Special's books is \$105,623.04, resulting in a gain of \$10,800 on the constructive retirement of the bonds. Note that Peerless' purchase price reflects the current market interest rate of 12.992186% when the bonds have 18 payments left to maturity.



Year 2

Peerless amortization table

date	interest payment	interest expense	amort.	discount	carrying value
Dec 31, 20X1				5,177	94,823
Jun 30, 20X2	6000	6,160	160	5,017	94,983
Dec 31, 20X2	6000	6,170	170	4,847	95,153



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration—YEAR 2

Bond Liability Entries—20X2 (Special Foods)

June 30, 20X2

Interest Expense	5,809		
Bonds Payable	191		
	Cash		6,000

Semiannual payment of interest.

December 31, 20X2

Interest Expense	5,799		
Bonds Payable	201		
	Cash		6,000

Semiannual payment of interest.

Bond Investment Entries—20X2 (Peerless)

June 30, 20X2

Cash	6,000		
Investment in Special Foods Bonds	160		
	Interest Income		6,160

Record receipt of bond interest.

December 31, 20X2

Cash	6,000		
Investment in Special Foods Bonds	170		
	Interest Income		6,170

Record receipt of bond interest.



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration—YEAR 2

◆ Subsequent recognition of gain on constructive retirement

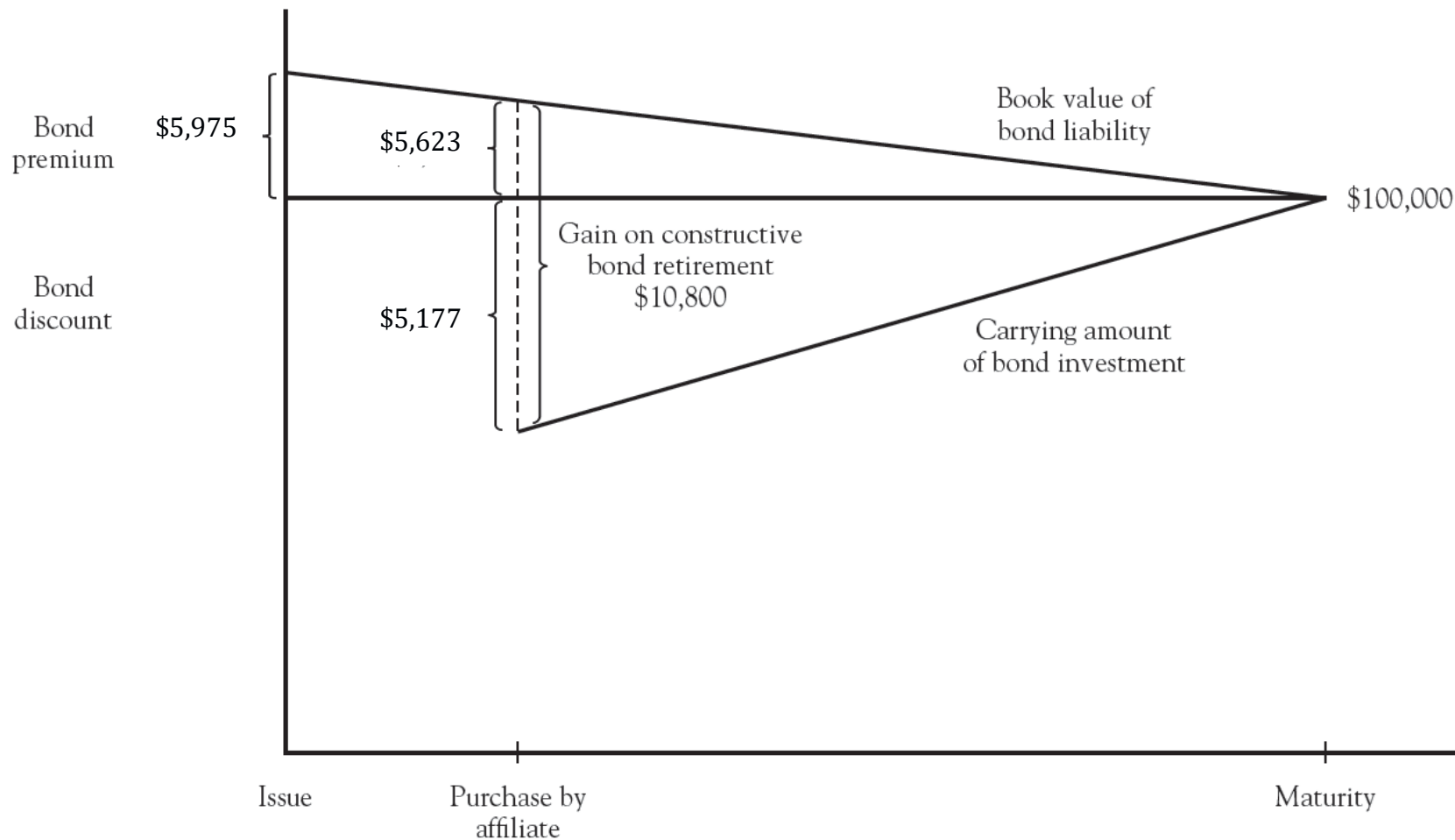
- In 20X1, the entire \$10,800 gain on the retirement was recognized in the consolidated income statement but not on the books of either Peerless or Special Foods.

In the subsequent years, both Peerless and Special Foods recognize a portion of the constructive gain as they amortize the discount on the bond investment and the premium on the bond liability (based on their respective amortization tables). Thus, the \$10,800 gain on constructive bond retirement, previously recognized in the consolidated income statement, is recognized on the books of Peerless and Special Foods over the remaining nine-year term of the bonds.



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration—YEAR 2

- This can be visualized as in the following figure:





Equity method entries—YEAR 2

Fully Adjusted Equity-Method Entries—20X2

Investment in Special Foods Stock	60,154	
	Income from Special Foods	60,154

Record Peerless' 80% share of Special Foods' 20X2 income.

Cash	32,000	
	Investment in Special Foods Stock	32,000

Record Peerless' 80% share of Special Foods' 20X2 dividend.

Income from Special Foods	578	
	Investment in Special Foods Stock	578

Recognize 80% share of amortization of premium and discount $[0.80 \times (191 + 201 + 160 + 170)]$

Whereas neither Peerless nor Special Foods recognized any of the gain from the constructive bond retirement on its separate books in 20X1, Peerless adjusts its equity method income from Special Foods for its 80 percent share of the \$10,800 gain, \$8,640. Therefore, as Peerless and Special Foods recognize the gain over the remaining term of the bonds, Peerless must reverse its 20X1 equity-method entry for its share of the gain. This adjustment is needed to avoid double-counting Peerless' share of the gain. Thus, the original adjustment of \$8,640 is reversed by the 80% portion of the combined amortization of Special Foods' premium and Peerless' discount each year.



Consolidation entries—YEAR 2

Book Value Calculations:

NCI 20%	Peerless 80%	Stock	+ Common Earnings	=	Retained +
Beginning Book Value	64,030	256,122		200,000	20,152
+ Net Income	3	60,154		,192	
– Dividends	(8,000)	(32,000)			
Ending Book Value	71,069	284,275		200,000	
155,344					

Basic Elimination Entry

Common Stock	200,000	← Common stock balance
Retained Earnings	120,052	← Beg. RE from trial balance
Income from Special Foods	59,576	← % of NI – 80% Gain Rec.
NCI in NI of Special Foods	14,894	← NCI % of NI – 20% 20X2 Gain Rec.
Dividends Declared	40,000	← 100% of Sub's dividends
Investment in Special Foods Stock		← Net BV – 80% 20X2 Gain Rec.
283,698*		← NCI % of BV – 20% Gain Rec.

* Rounding difference



Consolidation entries—YEAR 2

Item	Peerless Products	Special Foods	Unadjusted Totals	Consolidated Amounts
Bonds Payable	0	\$(105,231)	\$(105,231)	0
Investment in Bonds	\$95,153	0	95,153	0
Interest Expense	0	\$11,608	\$11,608	0
Interest Income	\$(12,330)	0	(12,330)	0

This analysis leads to the following worksheet entry to eliminate intercompany bond holdings.

Bonds Payable	105,231	
Interest Income	12,330	
Investment in Special Foods Bonds		95,153
Interest Expense	11,608	
Investment in Special Foods Stock		
NCI in NA of Special Foods		



Consolidation Worksheet at December 31, 20X2

	Peerless Products	Special Foods	Elimination Entries		Consolidated
			DR	CR	
Income Statement					
Sales	450,000	300,000			750,000
Less: COGS	(180,000)	(160,000)			(340,000)
Less: Depreciation Expense	(50,000)	(20,000)			(70,000)
Less: Other Expenses	(40,000)	(33,200)			(73,200)
Less: Interest Expense	(20,000)	(11,608)		11,608	(20,000)
Interest Income	12,330		12,330		0
Income from Special Foods	59,576		59,576		0
Consolidated Net Income	<u>231,906</u>	<u>75,192</u>	<u>71,906</u>	<u>11,608</u>	<u>246,800</u>
NCI in Net Income			<u>14,894</u>		<u>14,894</u>
Controlling Interest Net Income	<u>231,906</u>	<u>75,192</u>	<u>86,800</u>	<u>11,608</u>	<u>231,906</u>
Statement of Retained Earnings					
Beginning Balance	428,762	120,152	120,152		428,762
Net Income	231,906	75,192	86,800	11,608	231,906
Less: Dividends Declared	(60,000)	(40,000)		40,000	(60,000)
Ending Balance	<u>600,668</u>	<u>155,344</u>	<u>206,952</u>	<u>51,608</u>	<u>600,668</u>
Balance Sheet					
Cash	208,177	90,575			298,752
Accounts Receivable	150,000	80,000			230,000
Inventory	180,000	90,000			270,000
Investment in Special Foods Stock	292,338			283,698	0
				8,640	
Investment in Special Foods Bonds	95,153			95,153	0
Land	175,000	40,000			215,000
Building and Equipment	800,000	600,000		300,000	1,100,000
Less: Accumulated Depreciation	(500,000)	(340,000)	300,000		(540,000)
Total Assets	<u>1,400,668</u>	<u>560,575</u>	<u>300,000</u>	<u>687,491</u>	<u>1,573,752</u>
Accounts Payable	100,000	100,000			200,000
Bonds Payable	200,000	100,000	100,000		200,000
Premium on Bonds Payable		5,231	5,231		
Common Stock	500,000	200,000	200,000		500,000
Retained Earnings	600,668	155,344	206,952	51,608	600,668
NCI in NA of Special Foods				70,924	73,084
				2,160	
Total Liabilities & Equity	<u>1,400,668</u>	<u>560,575</u>	<u>512,183</u>	<u>124,692</u>	<u>1,573,752</u>



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration—YEAR 2

◆ Consolidated Net Income—20X2

Peerless' separate income		\$172,330
Special Foods' net income	\$75,192	
Peerless' amortization of bond discount	(330)	
Special Foods' amortization of bond premium	<u>(392)</u>	
Special Foods' realized net income		<u>74,470</u>
Consolidated net income, 20X1		\$246,800
Income to noncontrolling interest ($\$74,470 \times 0.20$)		<u>(14,894)</u>
Income to controlling interest		<u><u>\$231,906</u></u>



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration—YEAR 2

◆ Noncontrolling Interest—December 31, 20X2

Book value of Special Foods' net assets, December 31, 20X2:

Common stock		\$200,000
Retained earnings		<u>155,344</u>
Total book value of net assets		\$355,344
Gain on constructive retirement of bonds	\$10,800	
Less: Portion recognized by affiliates during 20X2	<u>(722)</u>	
Constructive gain not yet recognized by affiliates		<u>10,078</u>
Realized book value of Special Foods net assets		\$365,422
Noncontrolling stockholders' share		<u>× 0.20</u>
Noncontrolling interest in net assets, December 31, 20X2		<u>\$ 73,084</u>



Bonds of Affiliate Purchased from a Nonaffiliate: Illustration

- As of the beginning of 20X3, \$10,078 of the gain on the constructive retirement of the bonds remains unrecognized by the affiliates:

Gain on constructive retirement of bonds		\$10,800
Less: Portion recognized by affiliates during 20X2:		
Peerless' amortization of bond discount	\$330	
Special Foods' amortization of bond premium	<u>392</u>	
Total gain recognized by affiliates		<u>(722)</u>
Unrecognized gain on constructive retirement of bonds, January 1, 20X3		<u>\$10,078</u>



Consolidation entries in subsequent years

(1) Basic Elimination Entry

Common Stock
Retained Earnings
Income from Special Foods
NCI in NI of Special Foods
Dividends Declared
Investment in Special Foods Stock

- ← Common stock balance
- ← Beg. RE from trial balance
- ← % of NI – **OWN X Gain Rec.**
- ← NCI % of NI – **OWN X Gain Rec.**
- ← 100% of Sub's dividends
- ← Net BV – **OWN X Gain Rec.**
- ← NCI % of BV – **OWN X Gain Rec.**

(2) Bonds Elimination Entry

Bonds Payable table
Interest Income table

Investment in Special Foods Bonds table

Interest Expense table

Investment in Special Foods Stock OWN x

diff

NCI in NA of Special Foods OWN x diff

GAIN recognized = $\sum$ premium and discount amortized by parent and sub



Equity method entries—subsequent YEAR

Investment in Sub Stock
Income from Sub

Record Parent's share of Sub' income.

Cash	32,000
Investment in Sub Stock	32,000

Record Parent share of Sub' dividend.

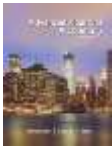
Income from Special Foods
Investment in Special Foods Stock

Recognize parent share of total amortization of premium and discount



Learning Objective 8-4

Prepare journal entries and elimination entries related to debt purchased from a nonaffiliate to an amount more than book value.



Bonds of Affiliate Purchased from a Nonaffiliate

- ◆ **Purchase at an amount greater than book value**
 - The consolidation procedures are virtually the same except that a loss is recognized on the constructive retirement of the debt



Bonds of Affiliate Purchased from a Nonaffiliate

Special Foods issues 10-year 12 percent bonds on January 1, 20X1, at par of \$100,000. The bonds are purchased from Special Foods by Nonaffiliated Corporation, which sells the bonds to Peerless on December 31, 20X1, for \$104,500. Special Foods recognizes \$12,000 ($\$100,000 \times .12$) of interest expense each year. Peerless recognizes interest income based on the following amortization table:

Bond Premium Amortization Table							
Payment Number	Period End	Interest Payment (Face $\times$ 0.12 $\times$ 6/12)	Interest Income (Carrying Value $\times$ 0.112393897 $\times$ 6/12)	Amortization of Premium	Premium	Bonds Payable Face Value	Carrying Value of Bonds
	12/31/20X1				4,500.00	100,000.00	104,500.00
1	6/30/20X2	6,000	5,848.75	(151.25)	4,348.75	100,000.00	104,348.75
2	12/31/20X2	6,000	5,840.28	(159.72)	4,189.03	100,000.00	104,189.03
3	6/30/20X3	6,000	5,831.34	(168.66)	4,020.37	100,000.00	104,020.37
4	12/31/20X3	6,000	5,821.90	(178.10)	3,842.27	100,000.00	103,842.27



Bonds of Affiliate Purchased from a Nonaffiliate

Because the bonds were issued at par, the carrying amount on Special Foods' books remains at \$100,000. Thus, once Peerless purchases the bonds from Nonaffiliated Corporation for \$104,500, a loss on the constructive retirement must be recognized in the consolidated income statement for \$4,500.

The bond elimination entry in the consolidation worksheet prepared at the end of 20X1 removes the bonds payable and the bond investment and recognizes the loss on the constructive retirement:

Eliminate intercorporate bond holdings:

Bonds Payable	100,000
Loss on Bond Retirement	4,500
Investment in Special Foods Bonds	
104,500	



Bonds of Affiliate Purchased from a Nonaffiliate

The bond elimination entry needed in the consolidation worksheet prepared at the end of 20X2 is as follows:

Eliminate intercorporate bond holdings:

Bonds Payable	100,000	
Interest Income	11,689	
Investment in Special Foods Stock	3,600	
NCI in NA of Special Foods	900	
Investment in Special Foods Bonds		104,189
Interest Expense	12,000	

$$\$11,689 = \$5,849 + \$5,840$$

$$\$3,600 = \$4,500 \times 0.80$$

$$\$900 = \$4,500 \times 0.20$$

$$\$12,000 = \$10,000 \times 0.12$$



Bonds of Affiliate Purchased from a Nonaffiliate

Similarly, the following entry is needed in the consolidation worksheet at the end of 20X3:

Eliminate intercorporate bond holdings:

Bonds Payable	100,000	
Interest Income	11,653	
Investment in Special Foods Stock		
NCI in NA of Special Foods		
Investment in Special Foods Bonds		103,842
Interest Expense	12,000	

$$\$11,653 = \$5,831 + \$5,822$$

$$\$3,451 = \$4,189 \times 0.80$$

$$\$838 = \$4,189 \times 0.20$$

$$\$103,842 = \$104,189 - \$169$$

- \$178

$$\$12,000 = \$100,000 \times 0.12$$



quiz

- ◆ Parent holds 80 percent of the voting shares of Sub Company. On January 1, 20X8, Sub purchased \$100,000 par value 12 percent Parent bonds from Third parties for \$115,000. Parent originally issued the bonds to Third Parties on January 1, 20X6, for \$110,000, assuming an 8.735 percent market rate. The bonds have an 8-year maturity from the date of issue and pay interest semiannually on June 30 and December 31 each year. Sub's reported net income of \$65,000 for 20X8, and Parent reported income (excluding income from ownership of Sub's stock) of \$90,000. Parent's partial bond amortization schedule is as follows:

PMT #		Interest \$ PMT	Interest Expense	Amort of Discount (Premium)	Premium (Discount)	Bonds Payable	BV of Bonds
	1/1/20X6				10,000.00	100,000.00	110,000.00
1	6/30/20X6	6,000.00	5,579.78	(420.22)	9,579.78	100,000.00	109,579.78
2	12/31/20X6	6,000.00	5,558.47	(441.53)	9,138.25	100,000.00	109,138.25
3	6/30/20X7	6,000.00	5,536.07	(463.93)	8,674.33	100,000.00	108,674.33
4	12/31/20X7	6,000.00	5,512.54	(487.46)	8,186.86	100,000.00	108,186.86
5	6/30/20X8	6,000.00	5,487.81	(512.19)	7,674.68	100,000.00	107,674.68
6	12/31/20X8	6,000.00	5,461.83	(538.17)	7,136.51	100,000.00	107,136.51



quiz

redemption Price		115,000
BV as Jan 1, 20X8		108,187
Loss		6,813



quiz

	Int received	Int Income	Amort premium	BV
		8.735%		115,000
30 Juni 20X8	6,000	5,023	977	114,023
31 Dec 20X8	6,000	4,980	1,020	113,003

Elimination entry			
Dr BP		107,137	
Dr Interest income		10,003	
Cr Loss on constructive retirement		6,813	
Cr Investment in Bond			(113,003)
Cr Interest expense			(10,950)



quiz



The End